



## Bloggers take notice

Google is about to ban all blogs that showcase X-rated content because of a sudden policy change. More here: [http://dgit.in/NSF\\_Ban](http://dgit.in/NSF_Ban)



## Better BoomSound

If a recent teaser released by HTC is to be believed, we can expect better BoomSound speakers soon: <http://dgit.in/BoomNew>

From the labs

# Med Sci: To V or not to V

The last few months have seen a surge in the controversy surrounding vaccination. Cases of measles have shot up in the US, with the blame going to parents who refuse to vaccinate their children for various reasons. We look at both sides of the story, and find out if there's any merit in not being vaccinated.

Daniel Dmello

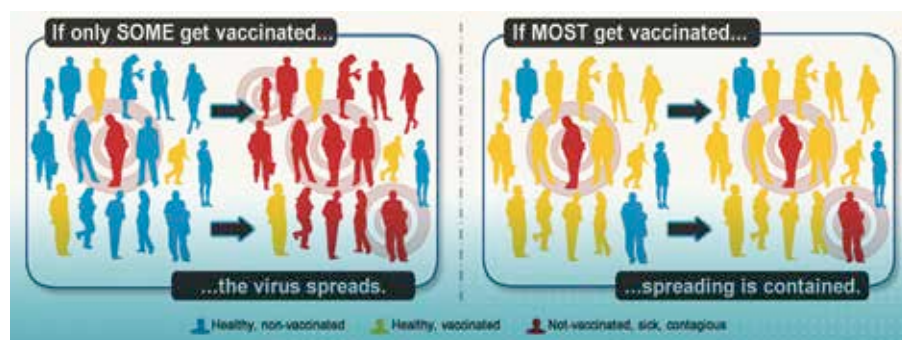
[readersletters@didgit.in](mailto:readersletters@didgit.in)

**T**he 52 kids who visited Disneyland, USA, in December 2014 and ended up contracting measles probably weren't prepared for such an eventuality. And neither were their parents. To a country like India, with a lot more measles cases, this doesn't seem serious. But to a country like the US, which had all but eradicated measles in 2000, this spurs fears of an epidemic. It also raises larger questions. Why did the children get measles? Why were they not vaccinated? Despite knowing what a serious disease measles is, many parents in the US refuse to vaccinate their kids, for fears of adverse effects. But how valid are these fears, and do they justify the risks of not being vaccinated? To answer this question, let's first look at some facts about vaccination.

## What is vaccination and why is it good?

There are dangerous viruses out there looking for host bodies in which they can multiply. Measles, for example, is a respiratory disease caused by one such virus. You can get it from someone else when they cough, sneeze or talk to you. Once you have the virus, your immune system becomes weakened and even a common cold can lead to death.

Diseases like these have no cure. If you get them, your body is on its own. Doctors can only try to increase your immunity to the disease. This is why they inoculate



The guy in the bottom right is really unlucky.

or vaccinate people. Vaccination involves giving children very small doses of the virus. This dose, consisting of weak or inactive parts of the virus, spurs the body into creating antibodies that it didn't previously have to fight off the disease. Now that the body has developed this form of self-defense, it is ready to tackle a real infection later in life.

Children who aren't vaccinated and who contract these diseases can still fight it off, provided that they have clean surroundings, rest and no other infections. However, this isn't always the case. One or two children out of a thousand will develop encephalitis along with measles and die from the disease, even those living in developed countries. Globally around 500 people, mostly children, die everyday from measles, of which one third live in India. That is around 56,000 child deaths a year or around 150 per day.

Therefore, it is better to be vaccinated, since all the evidence so far points towards a drop in death rates from vaccination.

The most common vaccine is the MMR (Mumps, Measles and Rubella) vaccine. Children need two doses, one at least nine months old and the second one a few months later, since 15% of children do not get immunity after the first dose.

## Herd Immunity

But there is more. It's not enough that most children are vaccinated. The vaccination rate for some diseases like measles needs to be extremely high, or the disease will still spread among the rest of the population.

Here is how this works. A highly transmissible disease like measles spreads like wildfire through a population, especially in densely populated areas. A small group of vaccinated children, say 50%, will not stop the spread of the disease. The disease will not spread through them, but any of the unvaccinated children are sufficient to carry it to the rest of the population. This is what happened at Disneyland. It takes just one person to infect an entire unvaccinated population. Which is why a relatively

SOURCE - CDC